

Dawei (David) Liu

Research focus: LLM agents, dependency-aware retrieval, reliable execution, and multimodal systems.

📍 Greater Seattle Area | ✉ davidliu02k@gmail.com

🌐 davidliuk.github.io | 🐙 github.com/davidliuk | 🔗 linkedin.com/in/davidliuk

EDUCATION

University of Pennsylvania

M.S.E. in Computer & Information Science | **GPA: 3.90/4.00**

Aug 2024 - May 2026

Philadelphia, PA

Northeastern University

B.E. in Software Engineering | **Rank: 5/396** (record: Sep 11, 2023)

Sep 2020 - Jun 2024

Shenyang, China

RESEARCH & OPEN-SOURCE EXPERIENCE

Selected Agent-Infrastructure Projects

Remote | 2026 - Present

- **Graph-of-Skills (GoS) | Primary developer & experiment lead.** Built dependency-aware retrieval with hybrid semantic-lexical seeding, reverse-weighted Personalized PageRank, and budget-aware context loading.
- **GoS evaluation:** Increased reward by **25.6%** (+7.0 absolute points) and reduced total tokens by **56.7%** vs. full skill loading on 1,000-skill SkillsBench (GPT-5.2 Codex). Highest average reward among evaluated methods in all six model-benchmark blocks across SkillsBench and ALFWorld.
- **Dr. Claw | Core contributor.** Implemented Gemini/Codex providers, multi-session and isolated-project workflows, CLI/desktop access, and reliability fixes; shipped cross-platform releases.
- **COMFYCLAW | Equal-contribution coauthor.** Co-developed a typed-graph harness with stage-aware tools, invalid-edit rollback, visual repair, and skill evolution.

PROFESSIONAL EXPERIENCE

ByteDance Inc.

Jun 2026 - Present

Software Engineer, Commerce Ads

Seattle, WA

- **Agent operations:** Built LLM-driven incident-triage workflows for evidence-grounded fault localization, incident-state tracking, recovery analysis, and recurring reporting.
- **Inference infrastructure:** Built an Anthropic/OpenAI-compatible gateway for coding agents, preserving streamed reasoning and tool continuation through context compression, concurrent turns, cancellation, and recovery.
- **Service validation:** Unified CI, test deployment, controlled traffic, and commit-traceable runtime verification in an AI-assisted workflow.

ByteDance Inc.

May 2025 - Dec 2025

Software Engineer Intern, Commerce Ads

Seattle, WA

- Developed Shop Ads image selection using posterior features, exploration-exploitation ranking, and multimodal LLM quality evaluation.
- Automated creative generation with scheduled generation and Flink-backed asset persistence for high-value products; built modular preview/diagnostics with test-entity injection for ad-format and rollout validation.
- Optimized product-serving reliability and tail latency through caching, asynchronous batch fetches, and offline computation.

Amazon

Jun 2024 - Aug 2024

Software Development Engineer Intern, Global-Mile Team

Beijing, China

- Instrumented distributed requests without application changes using a ByteBuddy-based OpenTelemetry Java agent for payload capture and tracing.
- Correlated traces by business ID across partially instrumented services; dynamically selected Kinesis or SQS for Fargate and Lambda runtimes.
- Built a React/Java diagnostics console with searchable trace, timeline, and payload views.

JD Technology

Jul 2023 - Oct 2023

Software Engineer Intern, Algorithm Tools

Beijing, China

- Redesigned Kubernetes resource management with ConfigMaps, Informers, asynchronous updates, and sharded locking to reduce cold-start overhead.
- Modularized campaign-page generation with the Strategy pattern and generative AI; improved multi-tenant delivery with Argo Workflows, Argo CD, and Helm.

SELECTED PUBLICATIONS

- Graph-of-Skills: Dependency-Aware Structural Retrieval for Massive Agent Skills** 2026
Dawei Liu[†], Zongxia Li[†], Hongyang Du, Xiyang Wu, Shihang Gui, Yongbei Kuang, Lichao Sun.
EMNLP 2026 (Main Conference); ACM CAIS 2026 Workshop AgentSkills. arXiv:2604.05333. [paper] [code]
- COMFYCLAW: Self-Evolving Skill Harnesses for Image Generation Workflows** 2026
Zongxia Li[†], Dawei Liu^{*}, Fuxiao Liu, Yuhang Zhou, Xiyang Wu, et al.
ECCV 2026 Workshop MUCG. arXiv:2607.01709. [paper] [code]
- Dr. Claw: An AI Scientist Workspace for Vibe Research** 2026
Dingjie Song, Hanrong Zhang, Dawei Liu, Yixin Liu, Zongxia Li, Zhengqing Yuan, et al.
EMNLP 2026 (System Demonstrations). arXiv:2609.00365. [paper] [code]
- A Cookbook of 3D Vision: Data, Learning Paradigms, and Application** 2026
Hongyang Du^{*}, Zongxia Li^{*}, Dawei Liu^{*}, Runhao Li^{*}, Haoyuan Song, Qingyu Zhang, et al.
CVPR 2026 Workshop OpenSUN3D. pp. 8376-8390. [paper] [code]
- Multimodal Video Generation Models with Audio: Present and Future** 2026
Dawei Liu, Hongyang Du, Zongxia Li, Xiyang Wu, Lantao Yu, Jingxi Chen, et al.
ACM Multimedia 2026 Workshop JAV-CG. [paper] [code]
- TIMEDB: tumor immune micro-environment cell composition database with automatic analysis and interactive visualization** 2023
Xueying Wang^{*}, Lingxi Chen^{*}, Wei Liu^{*}, Yuanzheng Zhang, Dawei Liu, et al.
Nucleic Acids Research. 51(D1):D1417-D1424. [paper] [site]
Contribution: Developed the visualization framework and several visualization analyses. [team contribution]

^{*} Equal contribution; [†] Core Contribution, as designated by the cited papers.

SELECTED HONORS

- Winner, Apple Swift Student Challenge** 2026
Created **Aura Chef**, a touchless Swift playground with hand-pose recognition and voice commands; selected as one of 350 winners from 37 countries and regions.
- China Collegiate Computing Contest - Mobile Application Innovation** 2022
Team lead & sole software developer of *Trace Note* (iOS/AR); National Final Second Prize and Northeast Division Innovation Track First Prize.
- National Scholarship for Undergraduate Students** 2020-2021
Ministry of Education, China.

RESEARCH IMPACT & SERVICE

- Google Scholar.** 80+ citations across 6 works; h-index 3; i10-index 2 (Sep 2026).
- Graph-of-Skills.** 203 stars, 25 forks; evaluated, compared, or reused in **13 source-checked papers**.
- Dr. Claw.** 1,050 stars, 116 forks; **51 non-merge commits** to main (4th among contributors).
- Peer review.** ACM CAIS Workshop AgentSkills (2026): completed two manuscript reviews through OpenReview.

TECHNICAL SKILLS

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| Languages | Python, Go, Java, C/C++, JavaScript/TypeScript, SQL, Swift |
| Agent/ML | LLM agents, tool calling, retrieval/GraphRAG, multimodal evaluation, PyTorch |
| Systems/Data | Distributed services, MySQL, Redis, Elasticsearch, Kafka; Flink, Thrift/Protobuf, OpenTelemetry |
| Cloud/DevOps | Docker, Kubernetes, AWS, Helm, CI/CD, Grafana |